



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES	DATE	14 JUNE 2023
GRADE	11	MARKS	50
EXAMINER	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
NAME		DURATION	50 MINUTES
CLASS			

INSTRUCTIONS:

1. This paper consists of 10 pages- 8 pages of questions and 2 pages of additional writing space. Please check that your question paper is complete.
 2. Answer all questions **on the question paper**.
 3. Write in blue or black pen only.
 4. Read all questions carefully and answer in as much detail as possible.
 5. All diagrams should have a clear heading.
 6. It is in your own interest to write legibly and to present your work neatly.
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QUESTION	POSSIBLE MARK	MARK ACHIEVED
1	14	
2	21	
3	15	
TOTAL: 50		

Question 1:

- 1.1. Various possibilities are suggested as answers to the following questions. Indicate the correct answer by writing the correct letter from the corresponding question in the space provided in the table below: (4)

1.1.1.	1.1.2.	1.1.3.	1.1.4.

- 1.1.1. Which of the following receives blood from the efferent arteriole?

- A. The renal vein
- B. The glomerulus
- C. The afferent arteriole
- D. The peritubular capillaries

- 1.1.2. Which of the following is least likely to be present in the glomerular filtrate (the filtrate produced by the nephron before it enters the loop of Henle) of a healthy adult nephron?

- A. Amino acids
- B. Glucose
- C. Electrolytes
- D. Proteins

- 1.1.3. The kidneys filter and remove...

- A. urea
- B. excess water
- C. blood
- D. A and B only

- 1.1.4. Presence of which of the following in the urine, even in small amounts is indicative of a possible problem in the renal system of a patient?

- A. Glucose
- B. Creatinine
- C. Urea
- D. Sodium

LIFE SCIENCES- EXCRETION AND SKELETAL SYSTEM

1.2. Select the term in Column B that best matches the description in Column A.

Write the letter of the term in the space provided to the left of Column A.

(10)

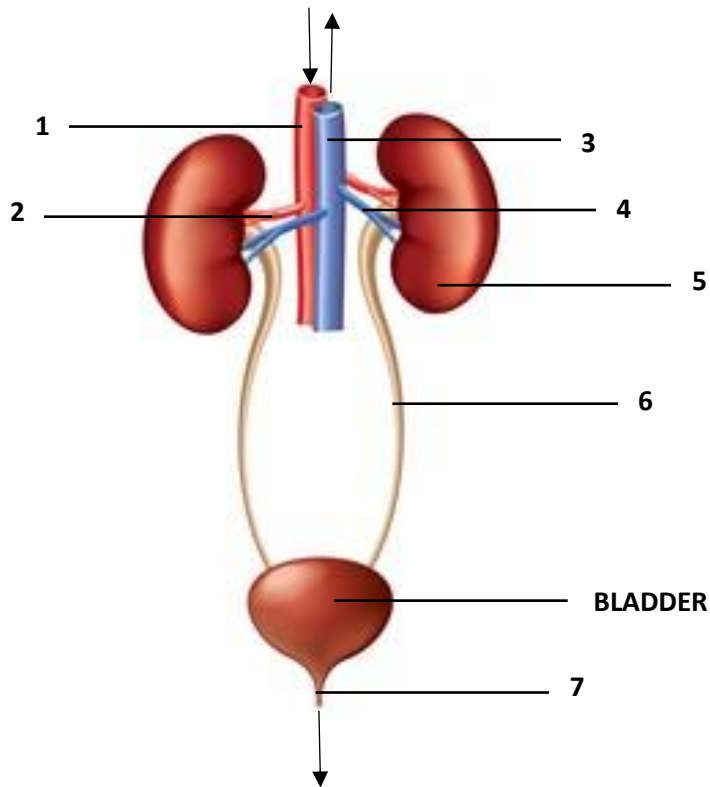
Column A**Column B**

- | | |
|---|-------------------------|
| [] The internal skeleton of vertebrates | A. Osteocytes |
| [] The part of the human skull that encloses the brain | B. Epiphysis |
| [] Bone cells | C. Periosteum |
| [] Connective tissue with a hard matrix | D. Cranium |
| [] The head of the femur | E. Vertebrae |
| [] The tissue covering the ends of long bones | F. Chondrocytes |
| [] Tough membrane that surrounds the bone | G. Endoskeleton |
| [] Irregular bone | H. Stress fracture |
| [] A crack in the bone that is caused by repeated pressure on the bone due to prolonged activities | I. Bone tissue |
| [] The skeleton of a squid | J. Greenstick fracture |
| | K. Exoskeleton |
| | L. Hyaline cartilage |
| | M. Hydrostatic skeleton |

[14]

Question 2

2.1. The diagram below shows the excretory system of a human being. Study the diagram and answer the questions that follow.



2.1.1. Provide labels for structures 1, 4, 6 and 7.

- 1- _____
4- _____
6- _____
7- _____

(4)

2.1.2. What is the value of the kidneys being surrounded by a thick layer of adipose tissue?

(2)

2.1.3. Tabulate two differences in the blood found in structures 2 and 4.

(4)

2.1.4. Explain why:

A. Urine is slightly thicker in summer than in winter.

(2)

B. The renal cortex has a dotted appearance.

(1)

2.2. Study the following table that shows the flow rate and concentration of certain substances taken at regions A, B, C and D of the nephron in the human kidney.

Part of nephron	Flow rate (cm ³ / min)	Solute concentrations (g/100 cm ³)				
		Proteins	Glucose	Sodium ions	Ammonium ions	Urea
A	4	0	0	0,6	0,04	1,80
B	200	0	0,10	0,72	0	0,05
C	40	0	0	0,3	0	0,15
D	2000	7	0,10	0,72	0	0,05

2.2.1. State and provide a reason which of the parts (A, B, C or D) of the nephron represent the following:

a. Afferent arteriole (2)

b. Bowman's capsule (2)

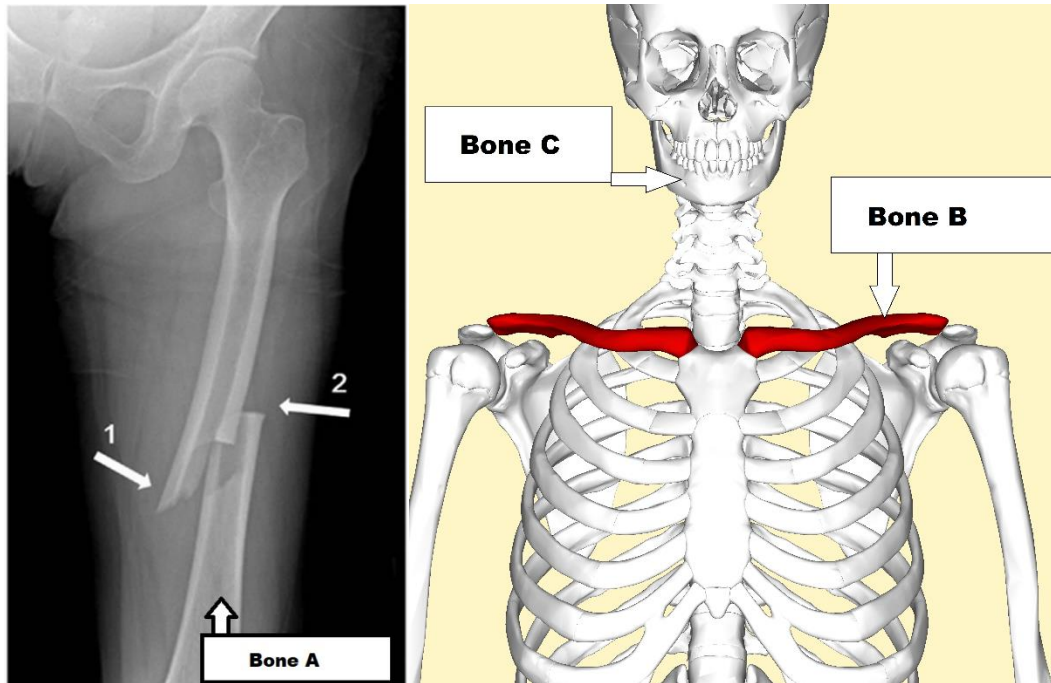
c. Loop of Henle (2)

d. Collecting duct (2)

[21]

Question 3

3.1. Look at the pictures below and answer the questions that follow.



An x-ray is a common diagnostic test that is used to detect injuries and diseases of the bones and joints. X-rays can determine a variety of bone abnormalities and diseases.

3.1.1. Provide the names of bones A, B and C. (3)

A- _____

B- _____

C- _____

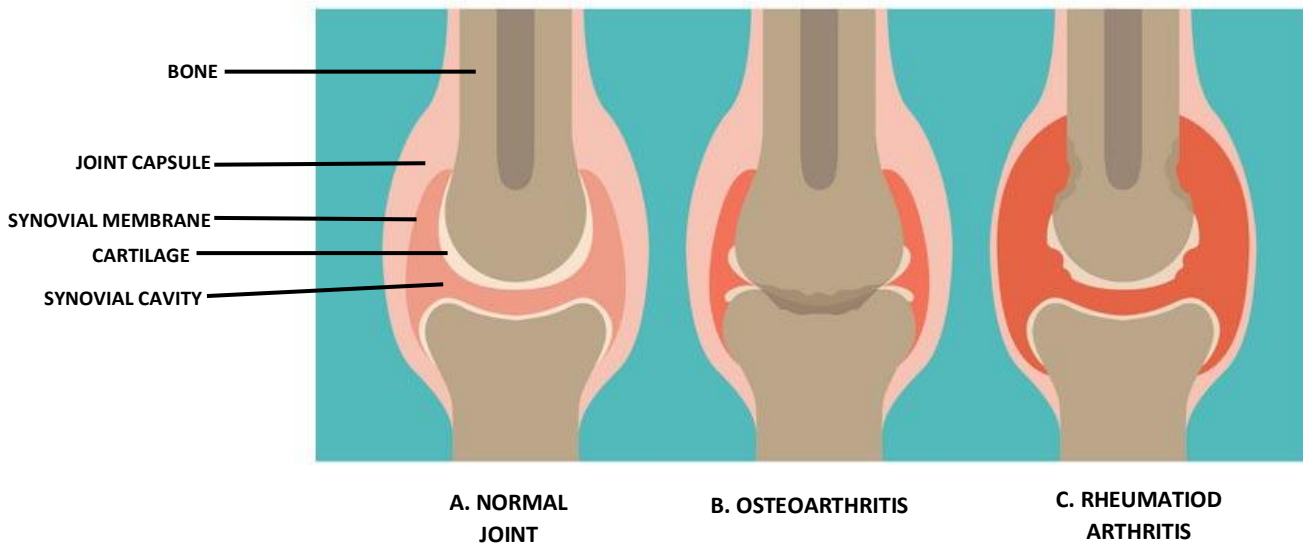
3.1.2. Which girdle of the body is shown in the x-ray above? (1)

3.1.3. What fracture is visible at 1 and 2 in the x-ray? (1)

3.1.4. Suggest what type of treatment this injury would require. (1)

3.1.5. Briefly describe the structure of the ribcage and to which part of the skeleton it belongs. (4)

3.2. Study the diagrams below and answer the questions that follow.



3.2.1. What type of joint classification is illustrated above? (1)

3.2.2. Provide one place in the body where the joint shown above can be found. (1)

3.2.3. What is osteoarthritis often referred to as? (1)

3.2.4. Differentiate between the conditions in B and C. Give visible differences. (2)

[15]

TOTAL: 50 MARKS

[illegible]

[illegible]